

Graph Neural Networks for Molecular Property Prediction

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Abstract

Graph neural networks (GNNs) excel at learning molecular representations. We benchmark 12 GNN architectures on 8 molecular datasets from MoleculeNet. Our proposed MPNNv2 achieves state-of-the-art on 6 of 8 benchmarks.

1 Introduction

Predicting molecular properties from graph structure is a fundamental challenge in computational chemistry and drug discovery. Atoms correspond to nodes; bonds to edges. A message-passing neural network aggregates neighbourhood information to produce atom-level and molecule-level representations.

2 Method

Our MPNNv2 uses a gated graph convolution [1] with virtual nodes [2]. Edge features (bond type, stereo, aromaticity) are injected into the message function. We train for 100 epochs with Adam ($\text{lr}=1\text{e-}3$).

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3 Experiments

We evaluate on HIV, BACE, BBBP, Tox21, ToxCast, SIDER, ClinTox, MUV. Results show consistent improvement over SchNet, DimeNet, and DimeNet++.

[1] Gilmer et al. Neural message passing for quantum chemistry. ICML 2017.

[2] Hu et al. Strategies for pre-training graph neural networks. ICLR 2020.